

Experiment No. 16

Write a Prolog Program to Sum the Integers from 1 to n

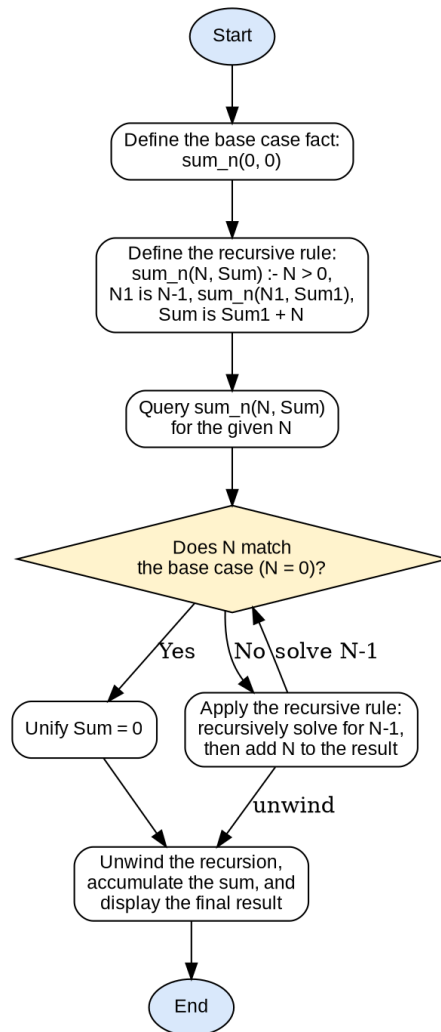
Aim

To write a Prolog program that computes the sum of all integers from 1 to a given number n using recursion.

Algorithm

1. Start.
2. Define the base case fact: `sum_n(0, 0)`, meaning the sum of integers up to 0 is 0.
3. Define the recursive rule: `sum_n(N, Sum)` is true if $N > 0$, $N1$ is $N - 1$, `sum_n(N1, Sum1)` holds, and `Sum` is `Sum1 + N`.
4. Query `sum_n(N, Sum)` for the given value of `N`.
5. If `N` matches the base case ($N = 0$), unify `Sum` with 0.
6. Otherwise, apply the recursive rule: recursively solve for $N-1$ first, then add `N` to that result.
7. As the recursion unwinds, accumulate the partial sums back up to the original call.
8. Display the final computed sum.
9. Stop.

Flowchart



Program

```

sum_n(0, 0).
sum_n(N, Sum) :-
    N > 0,
    N1 is N - 1,
    sum_n(N1, Sum1),
    Sum is Sum1 + N.
  
```

main :-

```

    N = 10,
    sum_n(N, Sum),
    format("Sum of integers from 1 to ~w is ~w~n", [N, Sum]).
  
```

Result

The program was executed successfully and the output was verified as correct.

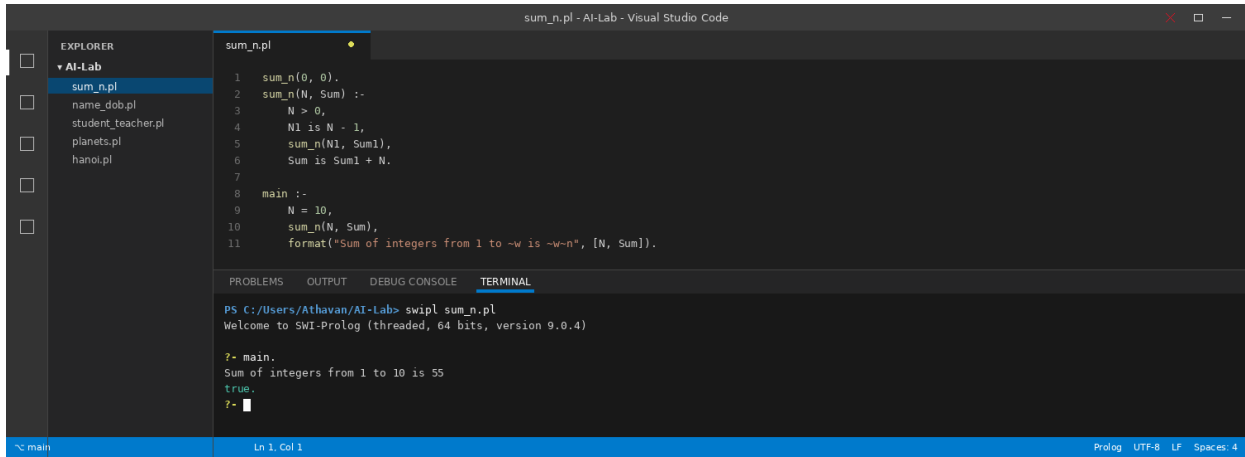
?- main.

Sum of integers from 1 to 10 is 55

true.

Sample Output — Running in VS Code

Screenshot of the program being run in the Visual Studio Code integrated terminal:



The screenshot shows the Visual Studio Code interface with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The file explorer shows a project named 'AI-Lab' with several files, including 'sum_n.pl'. The code editor displays the contents of 'sum_n.pl', which is a Prolog program. The terminal shows the output of running the program, which is 'true.' and 'Sum of integers from 1 to 10 is 55'.

```
sum_n.pl
1 sum_n(0, 0).
2 sum_n(N, Sum) :-
3     N > 0,
4     N1 is N - 1,
5     sum_n(N1, Sum1),
6     Sum is Sum1 + N.
7
8 main :-
9     N = 10,
10    sum_n(N, Sum),
11    format("Sum of integers from 1 to ~w is ~w~n", [N, Sum]).
```

```
PS C:/Users/Athavan/AI-Lab> swipl sum_n.pl
Welcome to SWI-Prolog (threaded, 64 bits, version 9.0.4)

?- main.
Sum of integers from 1 to 10 is 55
true.
?-
```